

NILESH KUMBHARE

AIML ENGINEER

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PUNE

PROFILE SUMMARY

AI / ML and Generative AI Engineer with **4+ years** of experience developing **Machine Learning, NLP, and LLM-based** solutions for real-world applications. Proficient in **Python and SQL, using Scikit-learn, TensorFlow, Keras, PyTorch, Pandas, and NumPy** for data processing and model development. Experienced in building LLM-powered systems using Retrieval-Augmented Generation (**RAG**), **Embeddings**, and **Prompt Engineering** for semantic search, question answering, and document intelligence. Skilled in **data preprocessing** and **feature engineering** to improve model performance and generalization, and in developing **classification, regression**, and recommendation models evaluated using **Precision, Recall, F1-score, ROC-AUC, and ranking metrics**, while working in **Agile environments** with focus on collaboration and technical documentation.

TECHNICAL SKILLS

- Generative AI & LLMs:** GPT-based Models, BERT, Sentence-BERT, Hugging Face, LangChain, OpenAI API, Retrieval-Augmented Generation (RAG), Prompt Engineering, RAG Pipelines, RAG Evaluation (RAGAS, BLEU, ROUGE)
- Agentic AI:** AI Agents, LangChain Agents, LangGraph, LangSmith, MCP
- Machine Learning:** Regression, Classification, Clustering, Hyperparameter Tuning, Cross Validation, Model Evaluation, Feature Engineering
- Deep Learning:** ANN, RNN, LSTM, GRU, Encoder-Decoder, Attention Mechanisms, Transformer Architectures, Transfer Learning
- Natural Language Processing (NLP):** TF-IDF, NER, Tokenization, Stemming & Lemmatization, Sentiment Analysis, Word2Vec, Embedding Techniques, Semantic Similarity
- Programming & Data Processing:** Python, SQL, Pandas, NumPy
- ML / DL Libraries:** Scikit-learn, TensorFlow, Keras, PyTorch, Matplotlib, Seaborn
- Vector Databases:** Pinecone, FAISS
- Deployment, MLOps & Cloud:** FastAPI, Docker, CI/CD Pipelines, AWS (S3, Lambda, EC2, SageMaker)
- Development & Collaboration:** Descriptive Statistics, Inferential Statistics, Hypothesis Testing, Scrum, Unit Testing, QA Migration, Git, GitHub, Jira

EDUCATION

- Gondwana University, Gadchiroli Bachelor of Science | 2020 - Summer

PROFESSIONAL EXPERIENCE

- ABCEDO Technology PVT LTD - Mumbai AIML Engineer | Sep 2021 - Present

ROLES & RESPONSIBILITIES :

Project : AI-Driven Quality & Compliance Document Intelligence Platform

- Designed and implemented an **LLM-assisted multi-RAG system** for semantic search and Q&A across regulated pharma documents.
 - Built separate **retrieval pipelines** for SOPs, QC reports, CAPA, and BMR/BPR using custom **chunking** and **metadata filtering**.
 - Developed **Sentence-BERT** embedding pipelines integrated with a Pinecone **vector database** for high-precision semantic retrieval.
 - Implemented **LangGraph orchestration** workflows for query routing, retrieval, validation, and controlled LLM generation.
 - Added **LLM guardrails** (rule validation, source grounding, confidence checks) to minimize **hallucinations** and ensure compliance.
 - Optimized LLM response quality through **prompt engineering**, result **re-ranking**, and **fallback retrieval** strategies.
 - Contributed to **deployment** using **FastAPI**, **Docker**, and **GitHub Actions**, ensuring scalability, seamless integration, and minimal manual intervention.
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Project : AI-Powered Personalized Product Recommendation System

- Designed a **personalized recommendation system** to improve product discovery, relevance, and user engagement on a **digital commerce platform**.
 - Analyzed **user interaction data** (click behavior, purchase history, product metadata) to derive preference signals for recommendations.
 - Implemented collaborative filtering and content-based techniques for candidate generation, followed by **ML-based ranking** and **re-scoring** using **Logistic Regression** and **XGBoost**.
 - Evaluated recommendation quality using **Precision@K** and **Recall@K**, ensuring alignment with engagement objectives.
 - Integrated **LLM-assisted** recommendation explanations and natural-language query handling, while keeping core recommendation **logic ML-driven**.
 - Deployed the solution as a **Flask REST API**, enabling **real-time review scoring** and integration with internal dashboards for actionable insights.
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Project : AI-Powered Personalized Product Recommendation System

- Developed a **machine learning-based churn prediction system** to identify telecom customers likely to discontinue services.
 - Performed **EDA** and **feature engineering** on usage, billing, and service interaction data to identify churn indicators.
 - Built and optimized XGBoost **classification models** with class imbalance handling and **hyperparameter tuning**.
 - Evaluated model performance using **Confusion Matrix**, **Precision**, **Recall**, and **ROC-AUC with cross-validation**.
 - Analyzed feature importance to **identify key churn drivers** and support retention strategies.
 - Delivered data-driven insights through **visualizations and model outputs** for decision-making.
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